

Laboratório 5

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Preparos

Pacotes

```
library(tidyverse)
```

Warning: pacote 'ggplot2' foi compilado no R versão 4.5.3

Warning: pacote 'dplyr' foi compilado no R versão 4.5.3

Warning: pacote 'forcats' foi compilado no R versão 4.5.3

Warning: pacote 'lubridate' foi compilado no R versão 4.5.2

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
v dplyr      1.2.1      v readr      2.1.5
```

```
v forcats   1.0.1      v stringr    1.5.1
```

```
v ggplot2   4.0.3      v tibble     3.2.1
```

```
v lubridate 1.9.5      v tidyr      1.3.1
```

```
v purrr     1.0.4
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(leaflet)
```

Warning: pacote 'leaflet' foi compilado no R versão 4.5.3

```
library(ggcal)
```

Informações dos Aeroportos

```
airports <- read_csv("airports.csv",  
  col_types = cols_only(  
    IATA_CODE = col_character(),  
    LATITUDE = col_double(),  
    LONGITUDE = col_double()  
  ) )
```

Crie uma função

```
analisa_aeronave <- function(tail_number, arquivo){  
  getStats <- function(input, pos) {  
    input %>%  
      filter(`CANCELLED` == 0,  
        TAIL_NUMBER == tail_number)}  
  
  flights <- read_csv_chunked(arquivo,  
    callback = DataFrameCallback$new(getStats),  
    chunk_size = 1e5,  
    col_types = cols())  
  
  flights <- flights %>%  
    mutate(DATE = as.Date(paste(YEAR, MONTH, DAY, sep = "-")  
      )  
    ) %>%  
    arrange(DATE, DEPARTURE_TIME)  
  
  origin <- airports %>%  
    select(  
      ORIGIN_AIRPORT = IATA_CODE,  
      ORIGIN_LAT = LATITUDE,  
      ORIGIN_LON = LONGITUDE  
    )  
}
```

```

destination <- airports %>%
  select(
    DESTINATION_AIRPORT = IATA_CODE,
    DEST_LAT = LATITUDE,
    DEST_LON = LONGITUDE

  )

flights <- flights %>%
  left_join(origin, by = "ORIGIN_AIRPORT") %>%
  left_join(destination, by = "DESTINATION_AIRPORT")

mapa <- leaflet() %>%
  addTiles()

for (i in 1:nrow(flights)){

  mapa <- mapa %>%
    addPolylines(
      lng = c(
        flights$ORIGIN_LON[i],
        flights$DEST_LON[i]
      ),
      lat = c(
        flights$ORIGIN_LAT[i],
        flights$DEST_LAT[i]
      ),
      weight = 1
    )
}

print(mapa)

return(
  list(
    tabela = flights,
    grafico = mapa))
}

```

#Mapa

```

resultado <- analisa_aeronave(
  "N431WN", file.path("flights.csv.zip")
)

resultado$tabela

```

```

# A tibble: 1,926 x 36
   YEAR MONTH DAY DAY_OF_WEEK AIRLINE FLIGHT_NUMBER TAIL_NUMBER
   <dbl> <dbl> <dbl>      <dbl> <chr>          <dbl> <chr>
1  2015     1     1          4 WN             1762 N431WN
2  2015     1     1          4 WN             3564 N431WN
3  2015     1     1          4 WN             3564 N431WN
4  2015     1     1          4 WN             3564 N431WN
5  2015     1     2          5 WN             1793 N431WN
6  2015     1     2          5 WN             1793 N431WN
7  2015     1     2          5 WN             1793 N431WN
8  2015     1     2          5 WN             1793 N431WN
9  2015     1     2          5 WN             1793 N431WN
10 2015     1     2          5 WN             3144 N431WN
# i 1,916 more rows
# i 29 more variables: ORIGIN_AIRPORT <chr>, DESTINATION_AIRPORT <chr>,
#   SCHEDULED_DEPARTURE <chr>, DEPARTURE_TIME <chr>, DEPARTURE_DELAY <dbl>,
#   TAXI_OUT <dbl>, WHEELS_OFF <chr>, SCHEDULED_TIME <dbl>, ELAPSED_TIME <dbl>,
#   AIR_TIME <dbl>, DISTANCE <dbl>, WHEELS_ON <chr>, TAXI_IN <dbl>,
#   SCHEDULED_ARRIVAL <chr>, ARRIVAL_TIME <chr>, ARRIVAL_DELAY <dbl>,
#   DIVERTED <dbl>, CANCELLED <dbl>, CANCELLATION_REASON <chr>, ...

```

Compilação de Data/Hora:

```

Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()

```

```
[1] "2026-09-10 15:27:48 -03"
```